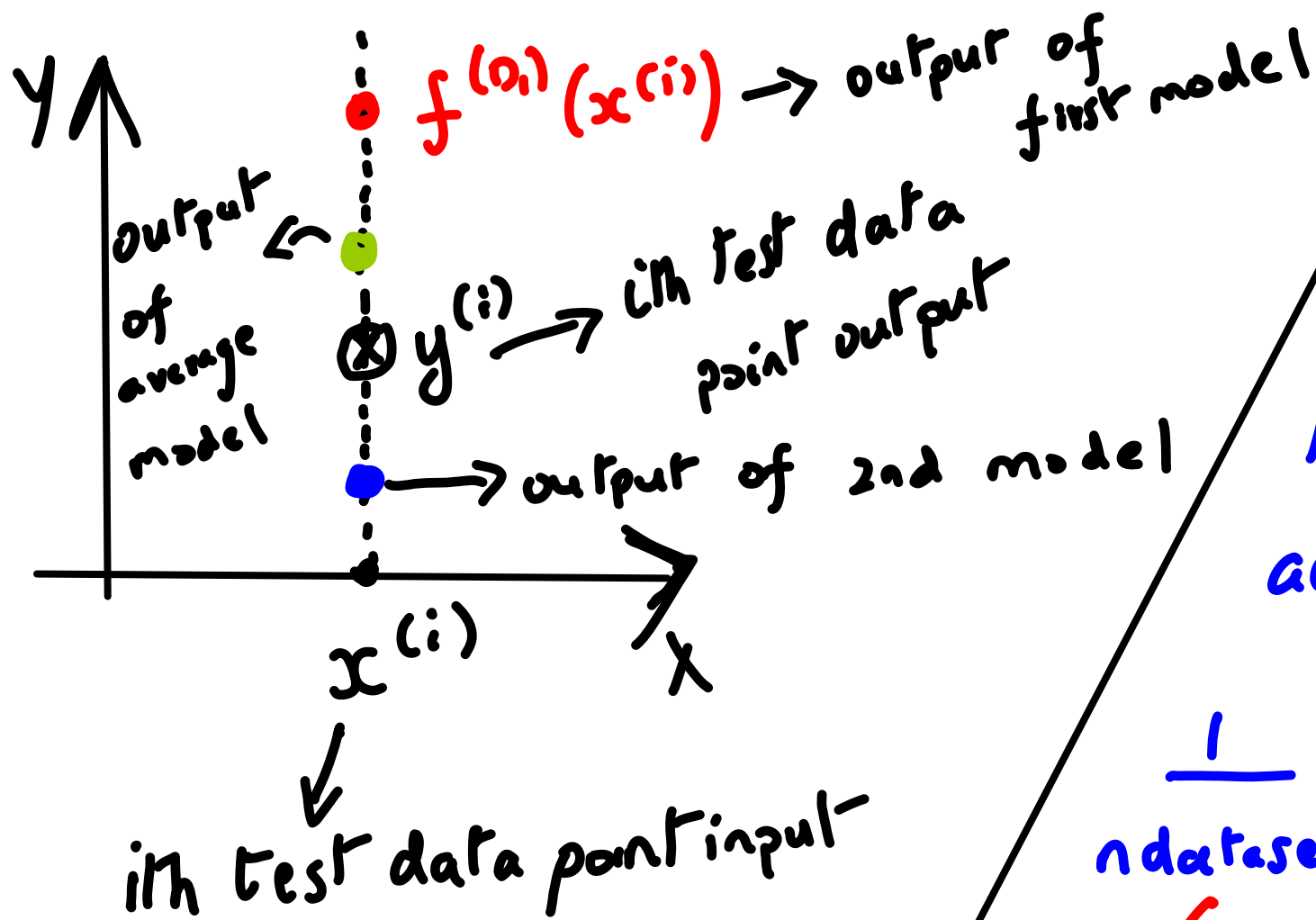




For the i th test data point, the mean squared error across all models is

$$\frac{1}{n_{\text{datasets}}} \sum_{k=1}^{n_{\text{datasets}}} \left[\underset{\substack{\downarrow \\ \text{Actual}}}{y^{(i)}} - \underset{\substack{\downarrow \\ \text{Kth model output}}}{f^{(D_k)}(x^{(i)})} \right]^2$$

The mean squared error can be split as:

$$\underbrace{\left[y^{(i)} - f^{\text{avg}}(x^{(i)}) \right]^2}_{\text{Average bias}} + \underbrace{\frac{1}{n_{\text{datasets}}} \sum_{i=1}^{n_{\text{datasets}}} \left[f^{(D_k)}(x^{(i)}) - f^{\text{avg}}(x^{(i)}) \right]^2}_{\text{Variance wr.t. average model}}$$